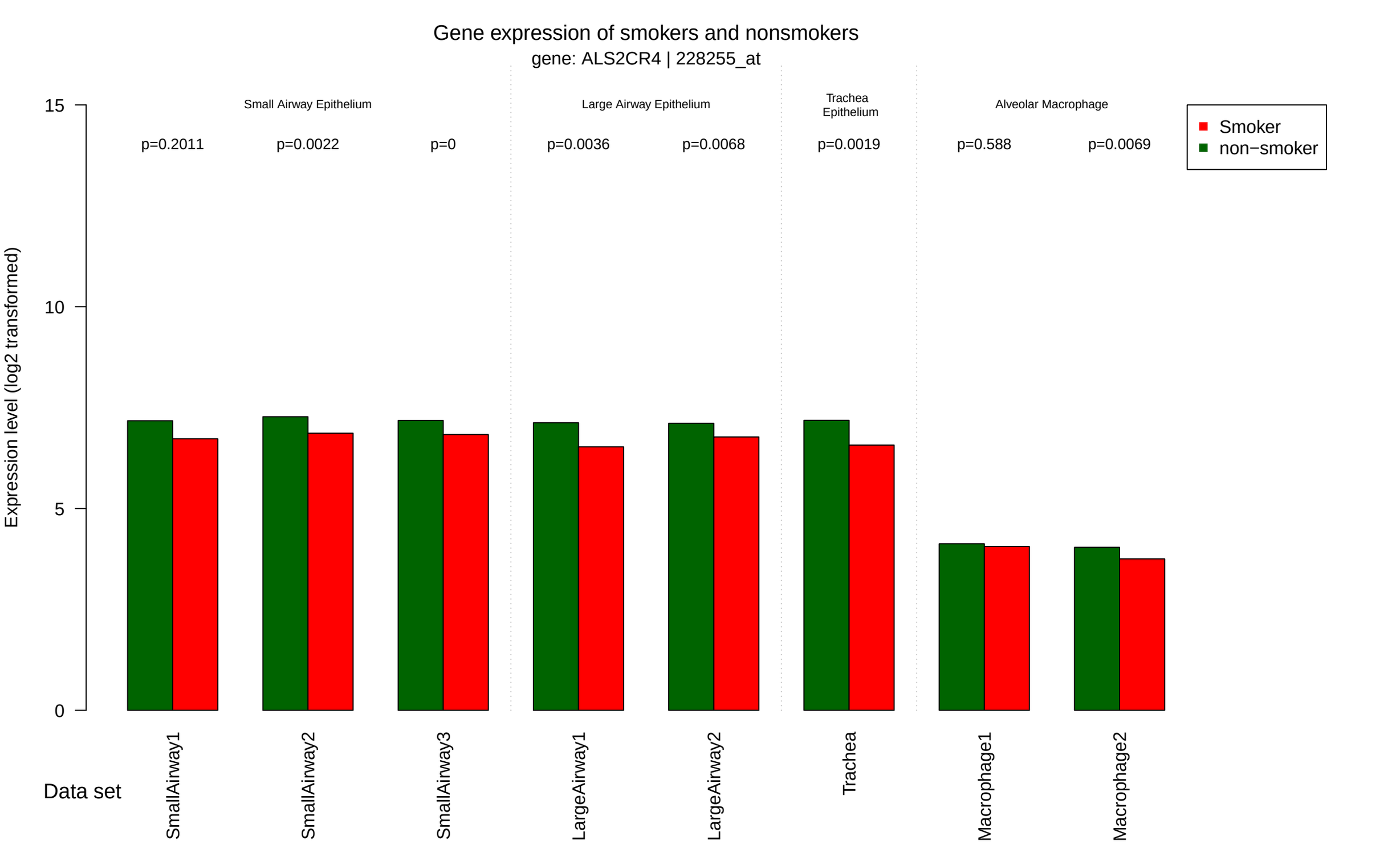
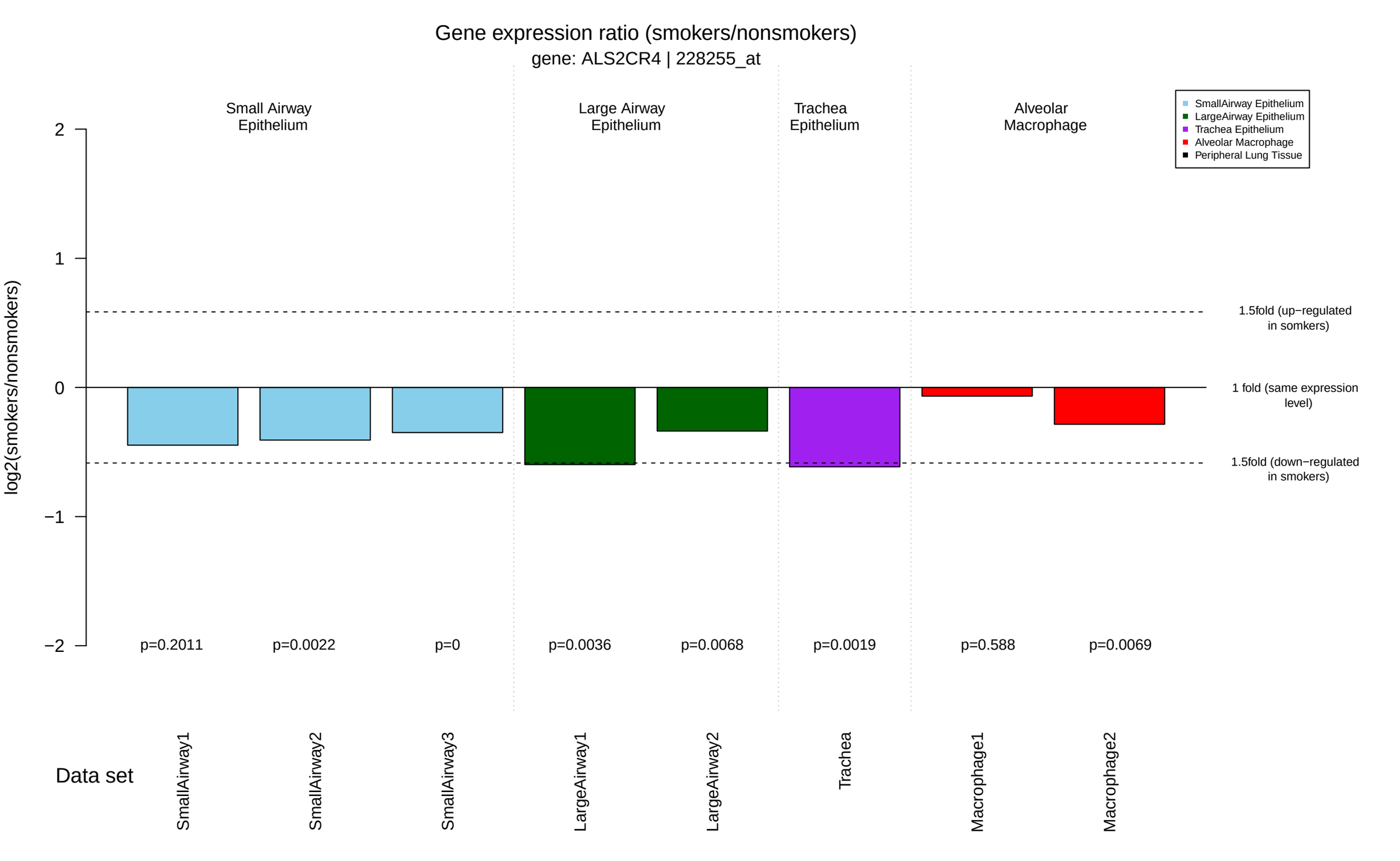
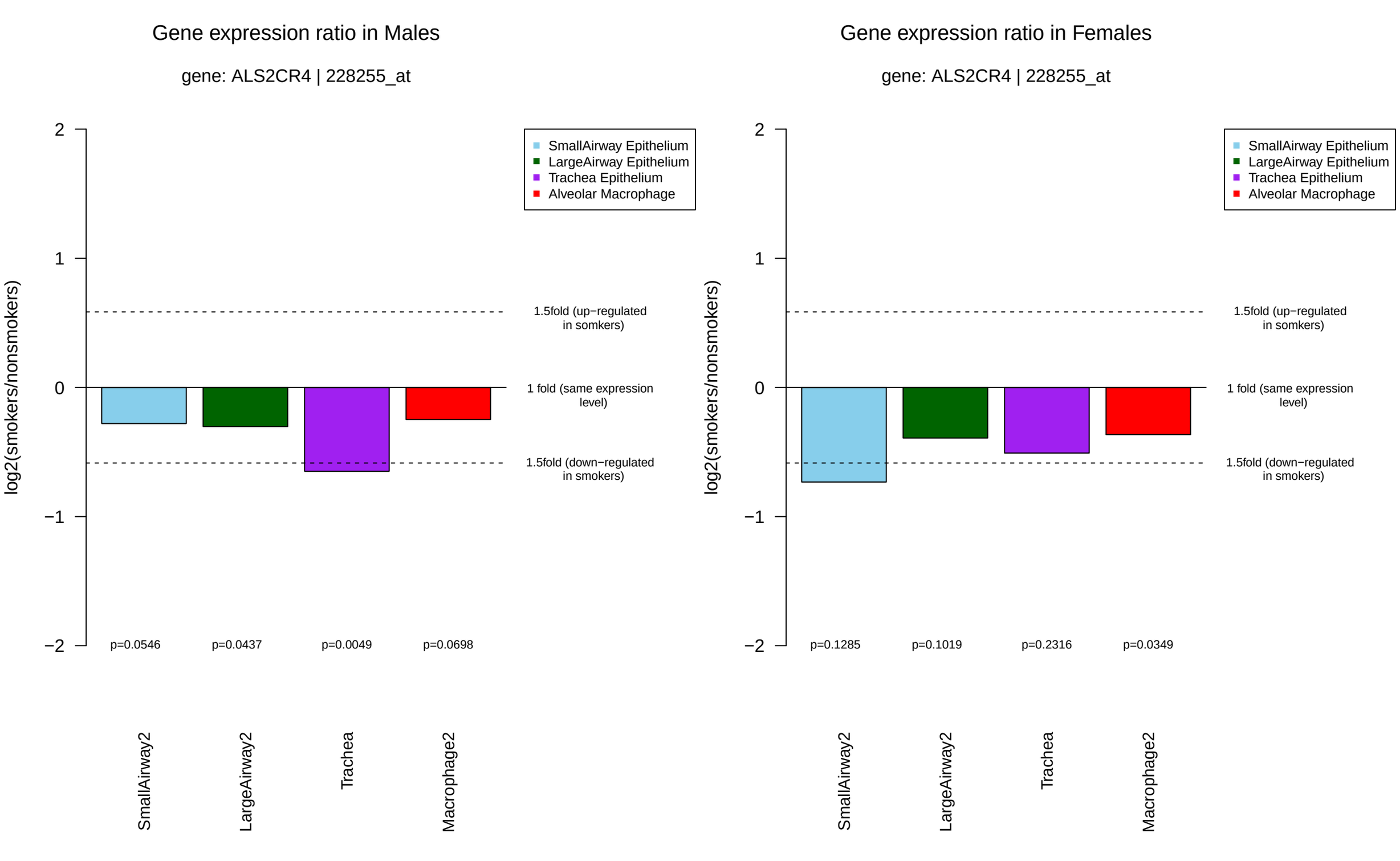
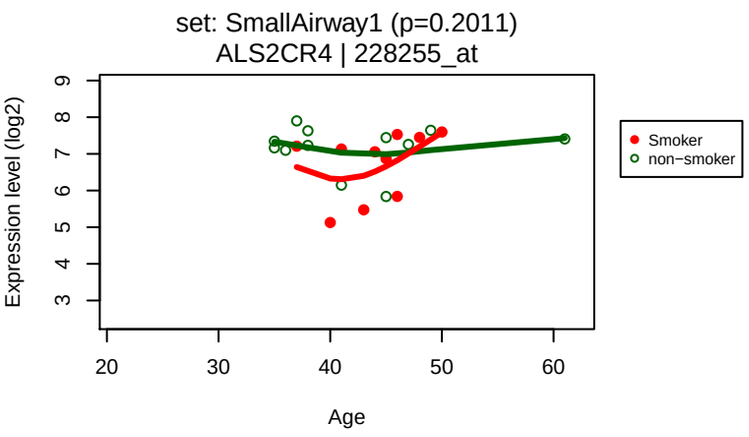
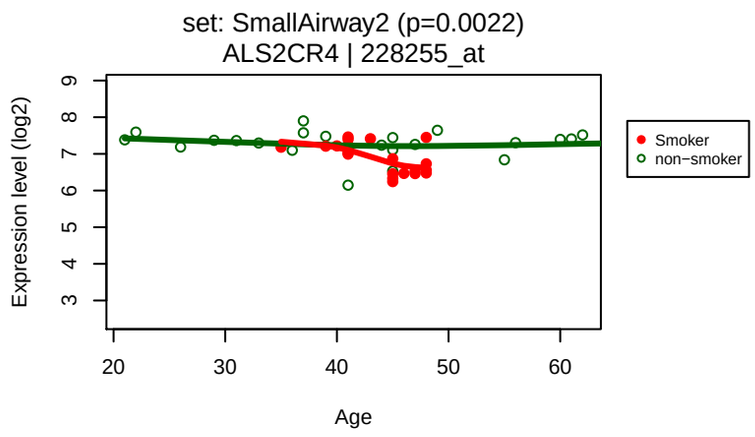
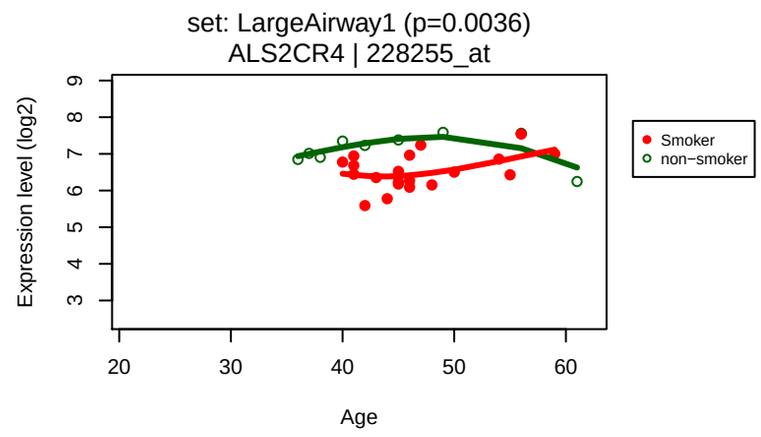

Gene expression of smokers and nonsmokers
gene: ALS2CR4 | 228255_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2011
p=0.0022
p=0
p=0.0036
p=0.0068
p=0.0019
p=0.588
p=0.0069
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

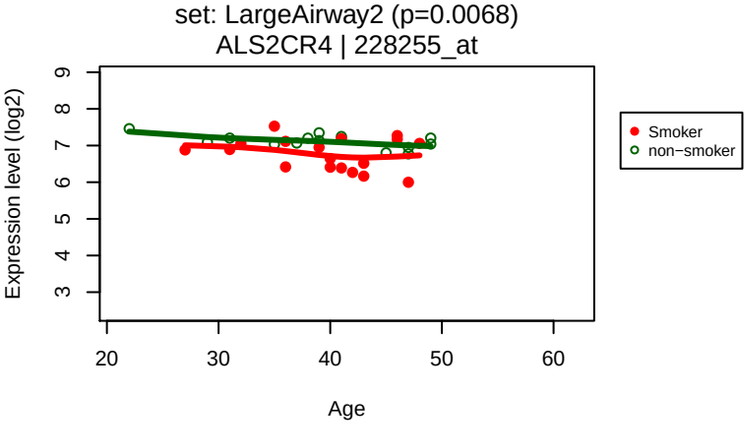

Gene expression ratio (smokers/nonsmokers)
gene: ALS2CR4 | 228255_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2011
p=0.0022
p=0
p=0.0036
p=0.0068
p=0.0019
p=0.588
p=0.0069
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

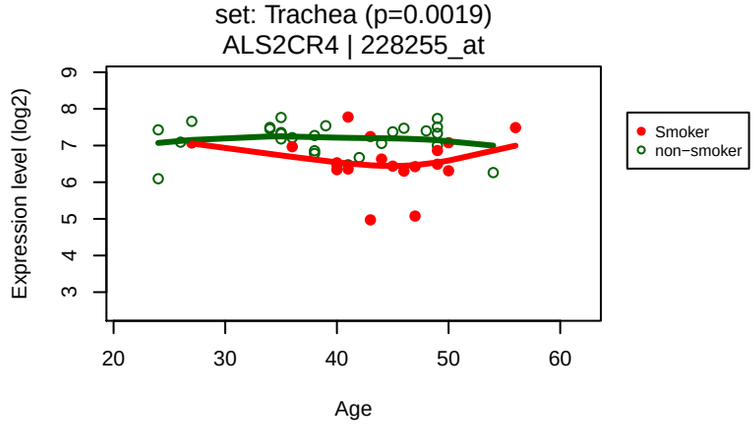

Gene expression ratio in Males
gene: ALS2CR4 | 228255_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0546
p=0.0437
p=0.0049
p=0.0698
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALS2CR4 | 228255_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1285
p=0.1019
p=0.2316
p=0.0349
SmallAirway2
LargeAirway2
Trachea
Macrophage2

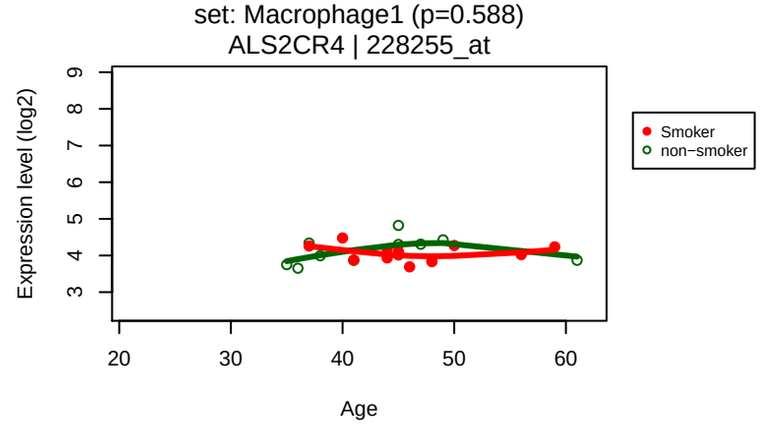

set: SmallAirway1 (p=0.2011)
ALS2CR4 | 228255_at
Expression level (log2)
Age
Smoker
non−smoker

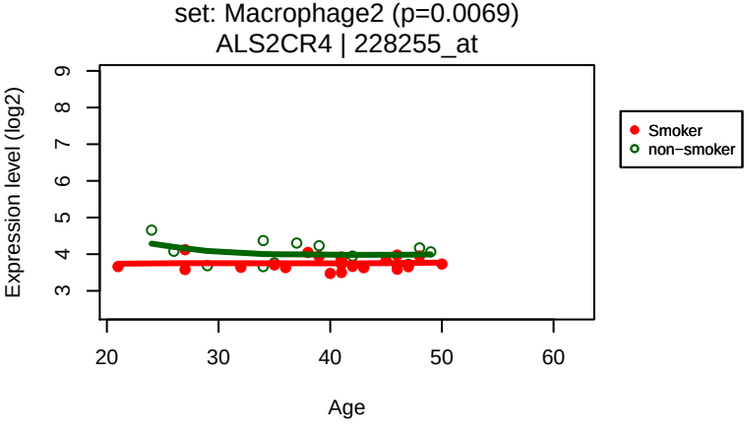

set: SmallAirway2 (p=0.0022)
ALS2CR4 | 228255_at
Expression level (log2)
Age
Smoker
non−smoker

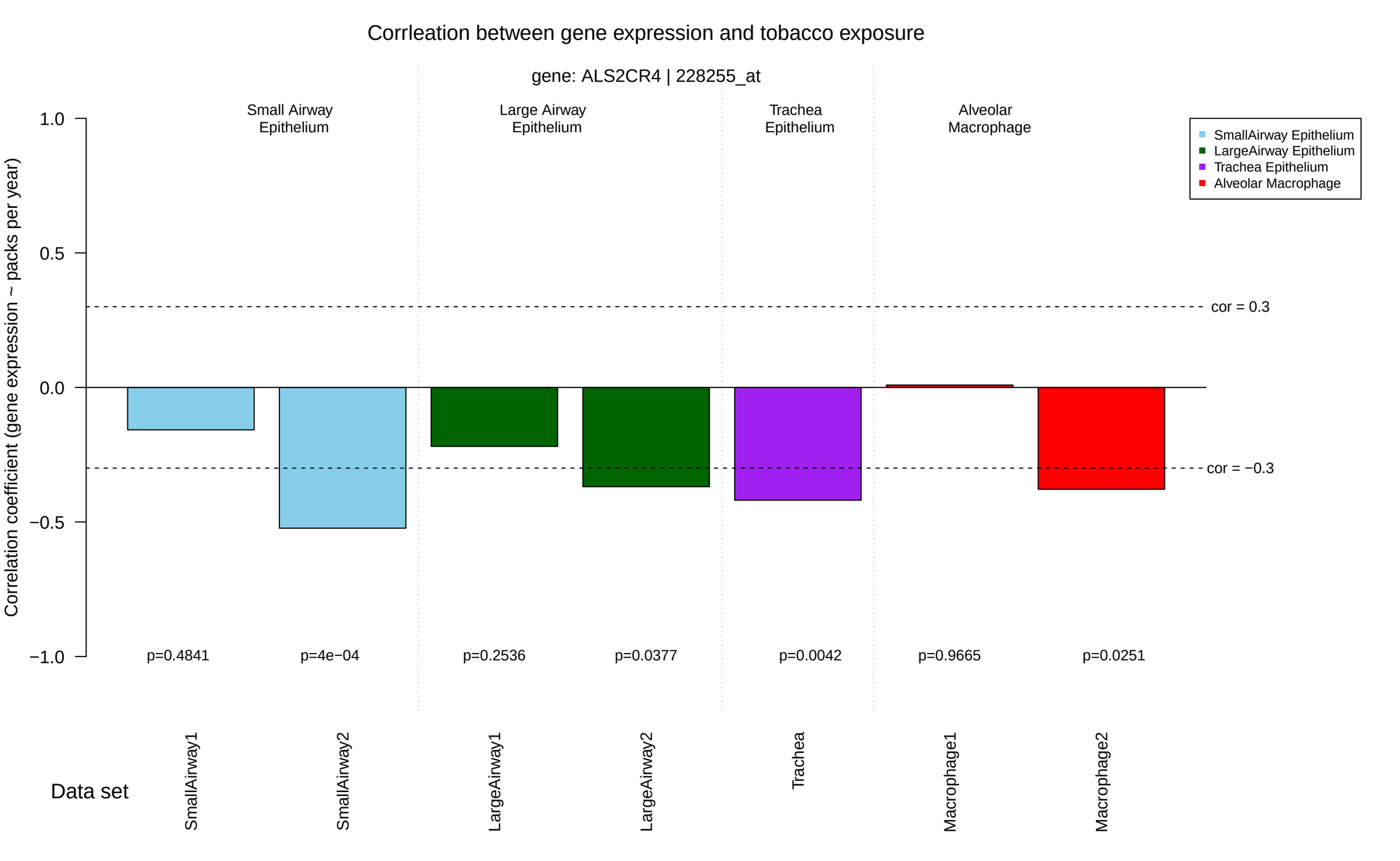

set: LargeAirway1 (p=0.0036)
ALS2CR4 | 228255_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0068)
ALS2CR4 | 228255_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0019)
ALS2CR4 | 228255_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.588)
ALS2CR4 | 228255_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0069)
ALS2CR4 | 228255_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALS2CR4 | 228255_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4841
p=4e−04
p=0.2536
p=0.0377
p=0.0042
p=0.9665
p=0.0251
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2